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Prediction of Gene Expression from Chromatin Landscape

Onepunch team

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Essential for any project

1. The data

- What do we have for this study/competition?
- What are the limitations of the current dataset?

Do Exploratory Data Analysis (EDA)!

2. Preprocessing

- What preprocessing steps are appropriate?

3. Feature engineering

- What features do we need and why?

4. Model selection

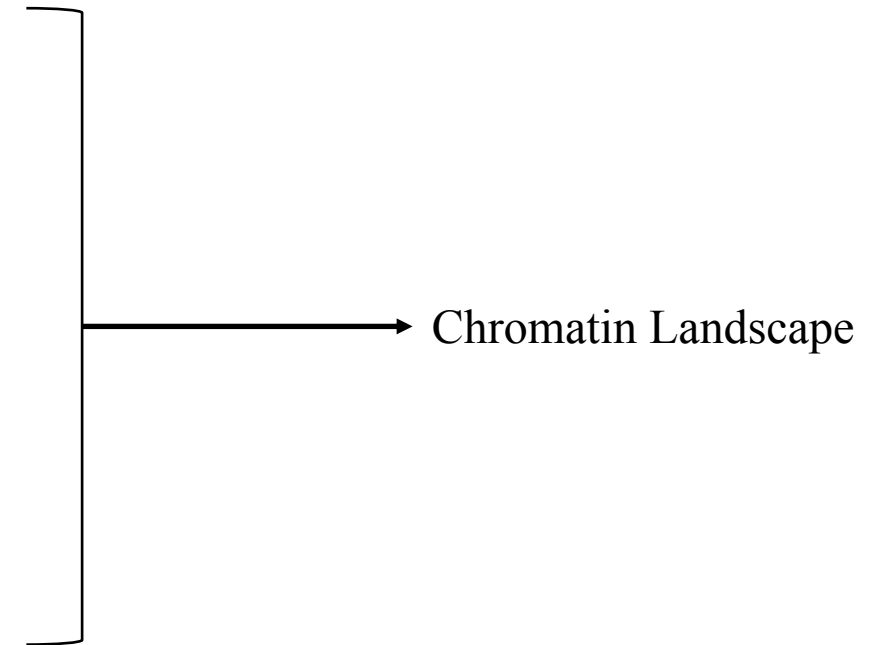
- Which state-of-the-art models fit this dataset?

5. Training & evaluation

- How will we design the training strategy and evaluation protocol?

Overview the dataset

- Gene Expression
- Gene Information
- H3K27me3 (ChIP-seq)
- H3K4me1 (ChIP-seq)
- H3K4me3 (ChIP-seq)
- H3K27ac (ChIP-seq)
- H3K36me3 (ChIP-seq)
- H3K9me3 (ChIP-seq)
- Chromatin Accessibility (DNase-seq)



EDA - Gene Information

Cell line: X1, X2, X3

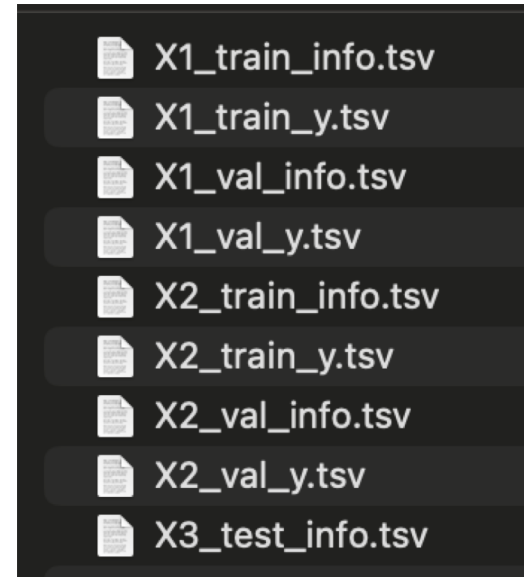
X1, X2 train = $\{chr2, chr3, chr4, chr5, chr6, chr7, chr8, chr9, chr10, chr11, chr12, chr13, chr15, chr16, chr17, chr18, chr20, chr21, chr22\}$

X1, X2 val = $\{chr14, chr19\}$

X3 test = $\{chr1\}$

This task is about cross-chromosome and cross-cell lines prediction!

Choose your training strategy wisely!



EDA – Reference Gene Information

X1 rows: 16284, X2 rows: 16284

❌ Differences found between X1 and X2 gene annotations.

The col: ['TSS_start', 'TSS_end'] makes the differences

⚠️ Number of differing entries: 2890

Sample differences (with renamed source flag):

	gene_name	chr	gene_start	gene_end	TSS_start	TSS_end	strand	_merge
3	A2ML1	chr12	8822621	8887001	8822471	8822521	+	from_X2
4	A2ML1	chr12	8822621	8887001	8844953	8845003	+	from_X1
5	A4GALT	chr22	42692121	42721298	42695583	42695633	-	from_X2
6	A4GALT	chr22	42692121	42721298	42721248	42721298	-	from_X1
15	AAMDC	chr11	77821109	77918432	77821108	77821158	+	from_X1
16	AAMDC	chr11	77821109	77918432	77821141	77821191	+	from_X2
26	AASS	chr7	122064583	122144308	122144205	122144255	-	from_X2
27	AASS	chr7	122064583	122144308	122144230	122144280	-	from_X1
33	ABCA12	chr2	214931542	215138626	215138354	215138404	-	from_X1
34	ABCA12	chr2	214931542	215138626	215138378	215138428	-	from_X2

- Only the TSS values differ.
- X1 and X2 share the same gene set.

Different TSSs can alter a gene's transcript abundance and isoform composition. ([Xi Wang et al. 2016](#))

[Karlsson et al. \(2017\)](#) showed that TSSs within the same gene are largely co-regulated, exhibiting positive correlations in usage.

$TSS_start = \min(TSS_start_X1, TSS_start_X2)$
 $TSS_end = \max(TSS_end_X1, TSS_end_X2)$

EDA – Reference Gene Information

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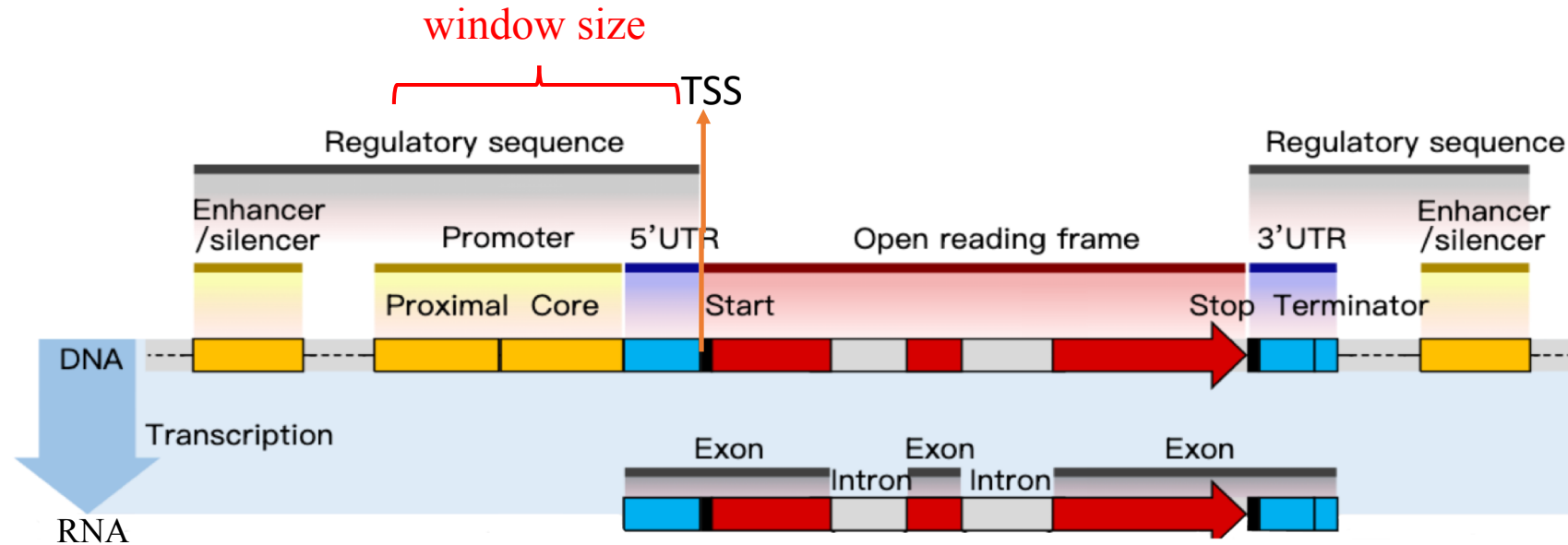
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- Only the TSS values differ.
- X1 and X2 share the same gene set.

Therefore, all three cell lines should share the same gene set.

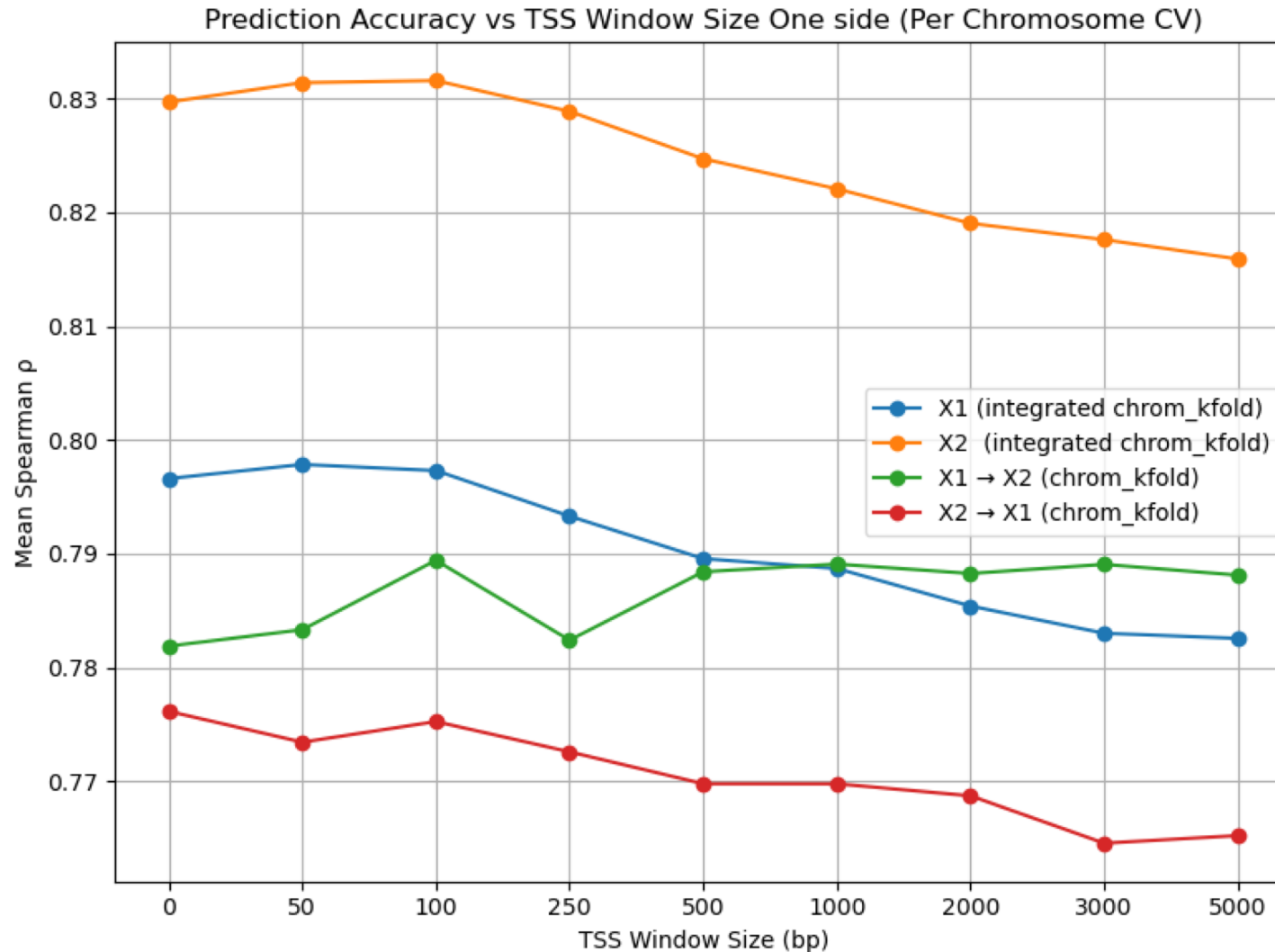
X1, X2 + X3 = reference chr1-22 gene set

EDA – Gene Information



Key regulatory signals cluster near the promoter/TSS ([ENCODE Project Consortium et al., 2020](#)). Therefore, we need to set and **find the optimal window size**.

EDA – The Optimal Window Size



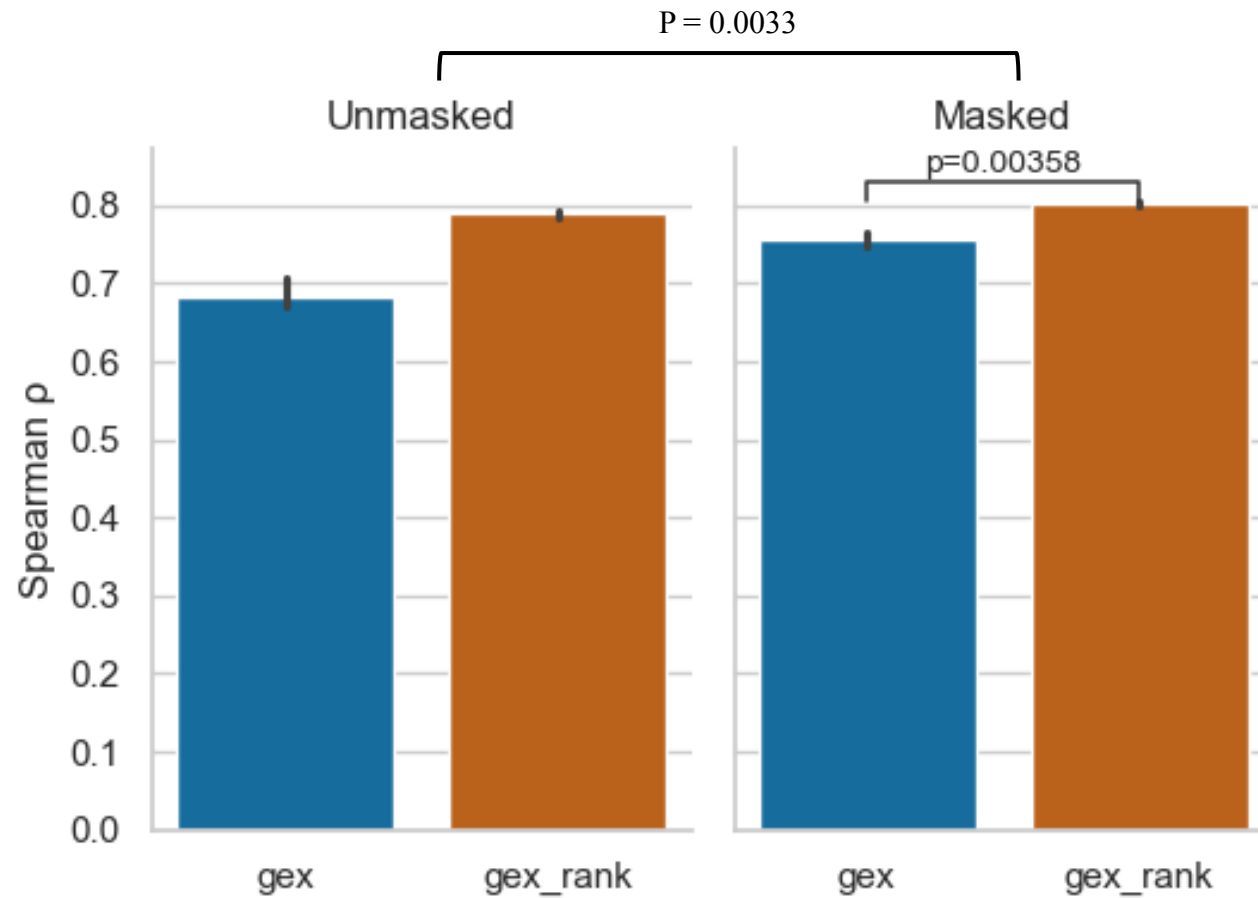
EDA – Gene Expression

```
=== X1 (train) | gex = 0: 9448 (58.02%) ===  
=== X1 (train) | gex > 0: 6836 (41.98%) ===  
mean      123.398521  
std       483.509682  
min       0.480536  
25%       2.114356  
50%       16.914850  
75%       83.997610  
max       19519.833527  
=== X2 (val) | gex = 0: 8078 (49.61%) ===  
=== X2 (val) | gex > 0: 8206 (50.39%) ===  
mean      105.014021  
std       461.589947  
min       0.096071  
25%       0.672496  
50%       7.877815  
75%       62.494134  
max       19512.484272
```

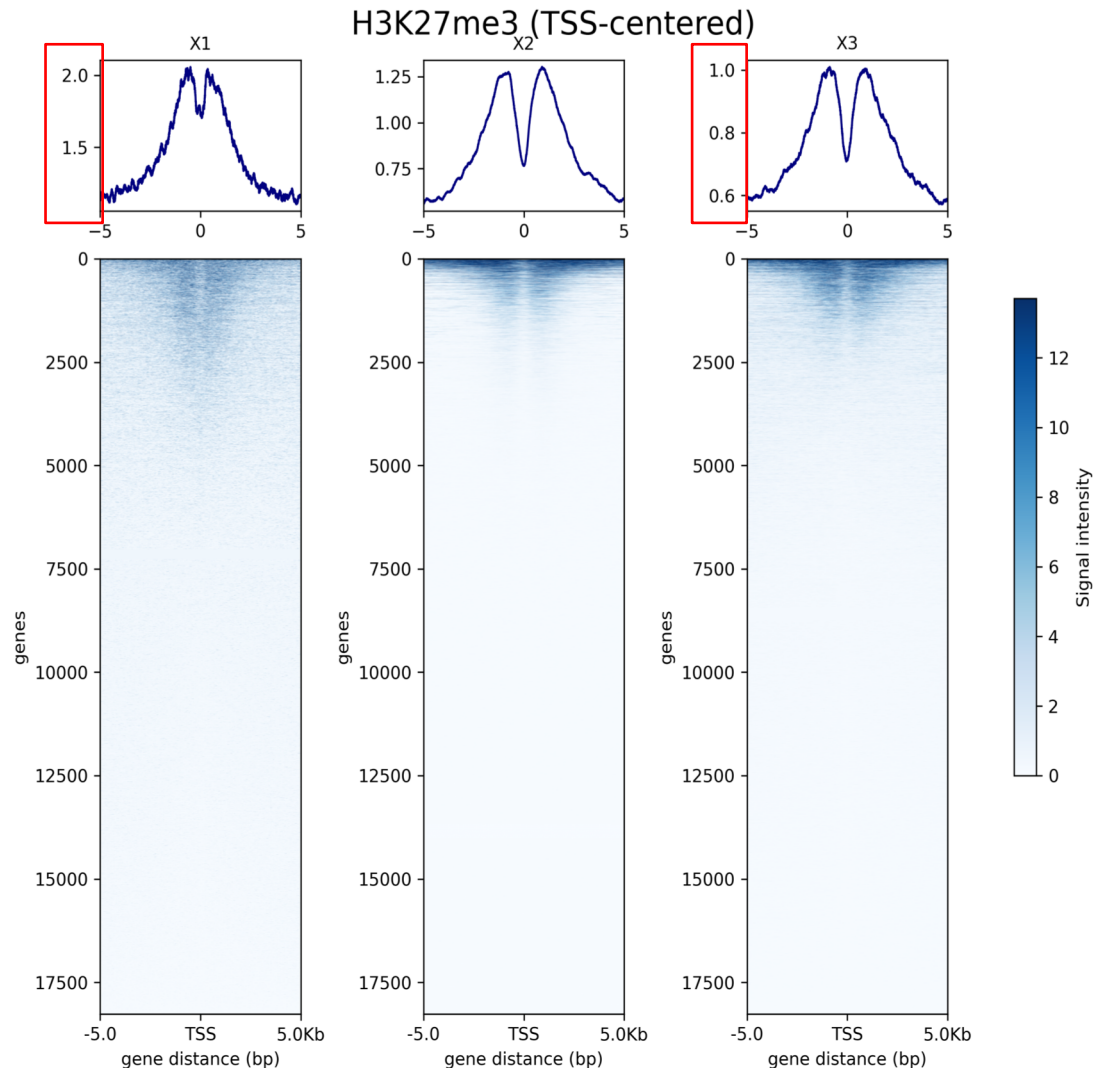
- Roughly 50% zeros
Consider a two-stage approach (classifier for zero vs. non-zero, then regression on positives).
- Skewed, long-tailed *gex*
Apply a transformation (e.g., log1p/Box-Cox/yeo-Johnson/rank) to stabilize gradients and improve learnability.

gex -> *rank_gex*

EDA – Gene Expression



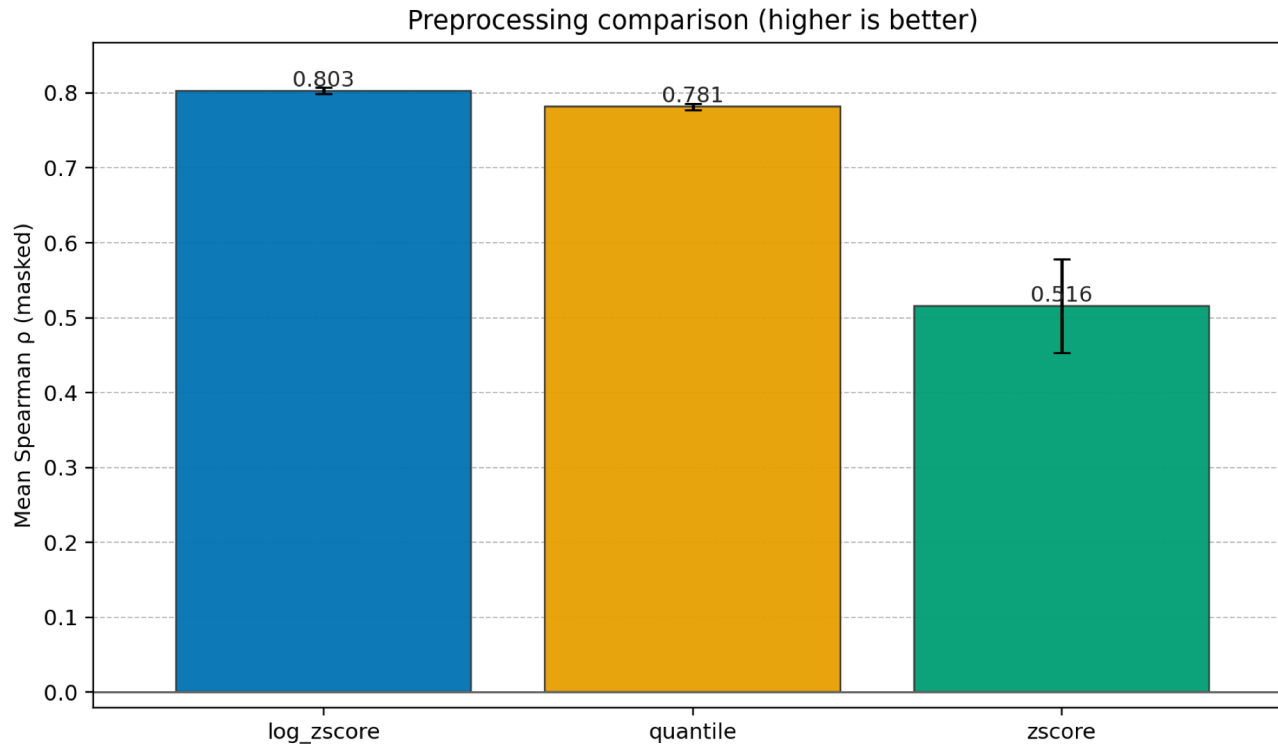
EDA – Chromatin Landscape



- Chromatin signal scales differ across cell lines, indicating potential batch effects.

Apply transformations (e.g., log1p, per-track z-scoring, or quantile normalization) to reduce batch effects and improve generalization.

EDA – Chromatin Landscape



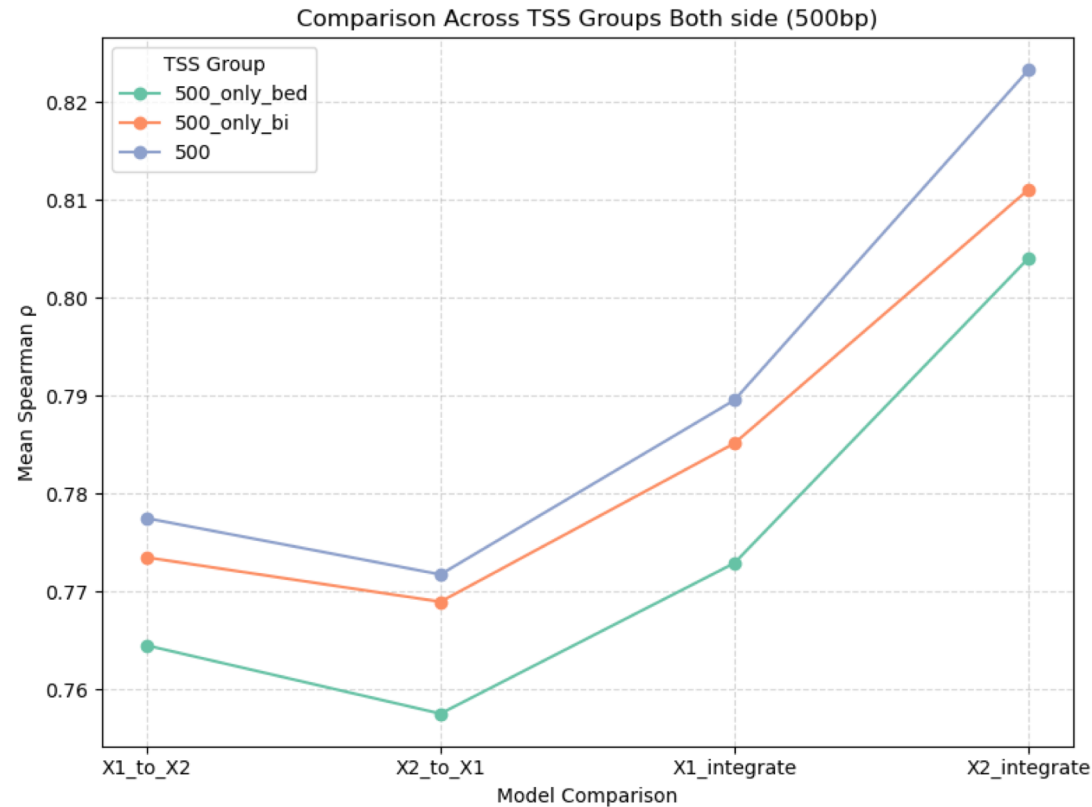
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log1p transform - > z-score

EDA – BigWig vs. BED

- A BigWig file is a continuous signal track that stores a value per genomic position.
- A BED file lists genomic regions with the mean signal value.



Both > BigWig > BED

Features Engineer

- **BigWig & BED features:** signal_sum, signal_mean, signal_std, signal_min, signal_max, signal_diff etc.



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 **Gene-expression-prediction** Public

This is the project 1 for the ML 4 genomics course in ETH. The project is to use the ChIP-seq dataset to predict the expression of the gene.

 Jupyter Notebook

Model selection

- Our data are tabular, and chromatin-derived features can exhibit non-linear interactions.
- I used **LightGBM** for both binary classification and regression because it
 - efficiently captures non-linearities and feature interactions,
 - trains fast and scales well,
 - is more interpretable than deep nets (feature importance/SHAP), and
 - handles heterogeneous features and missing values with minimal preprocessing.

The Training Strategy

This task is about cross-chromosome and cross-cell lines prediction!

- Cell line
- Chromosome

The Training Strategy - Cell line

Pooled training (mix all cell lines):

Train/validate on the combined dataset from all cell lines.

Within-cell line:

Train and validate on the same cell line (no cross-line mixing).

Cross-cell line:

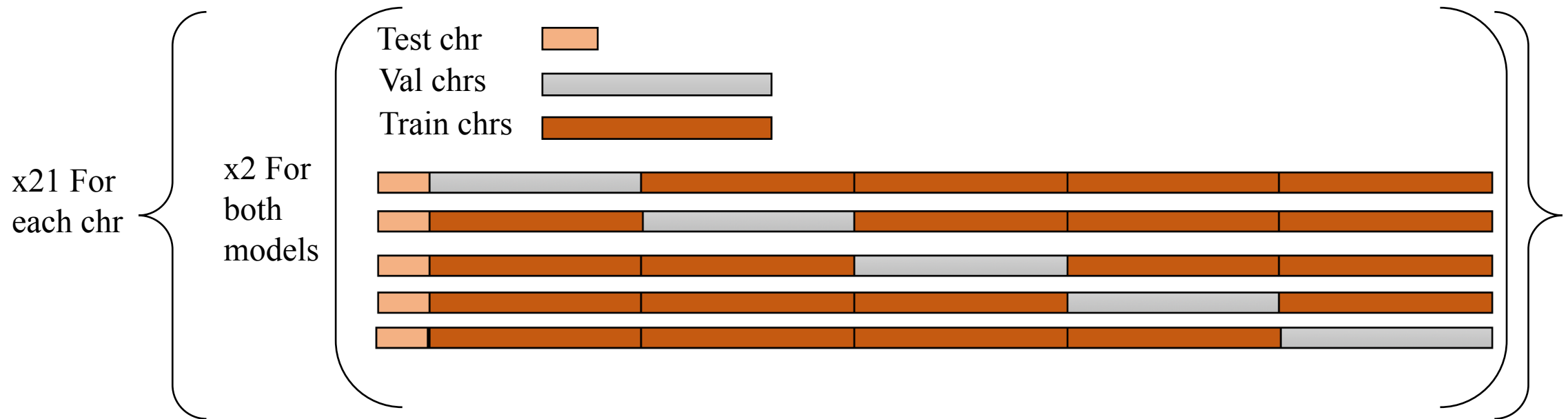
Train on one cell line and validate on a different cell line.

Pooled training: Captures the most general, cross-cell line patterns.

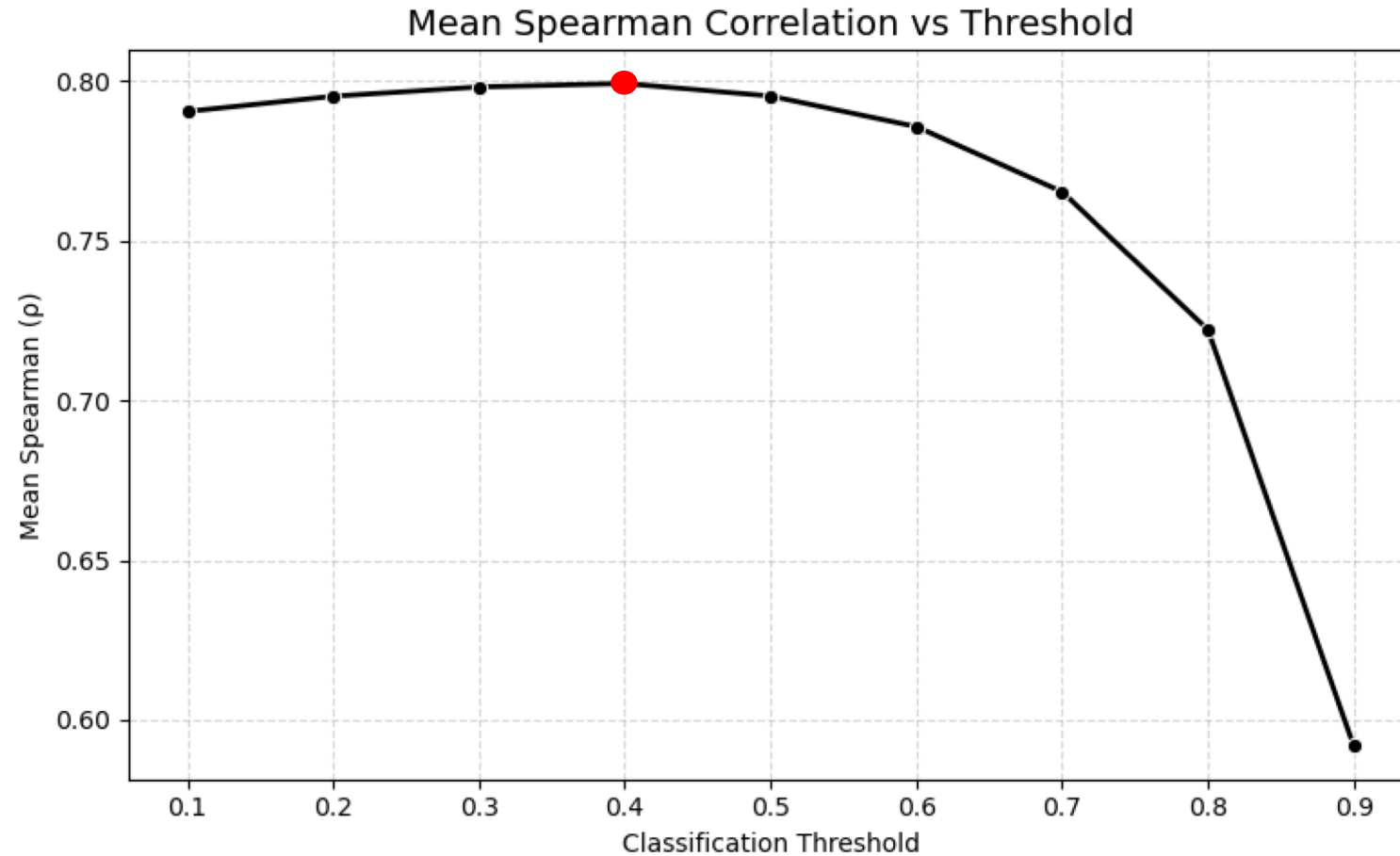
Within-cell-line training: Emphasizes cell line-specific patterns.

The Training Strategy - Chromosome

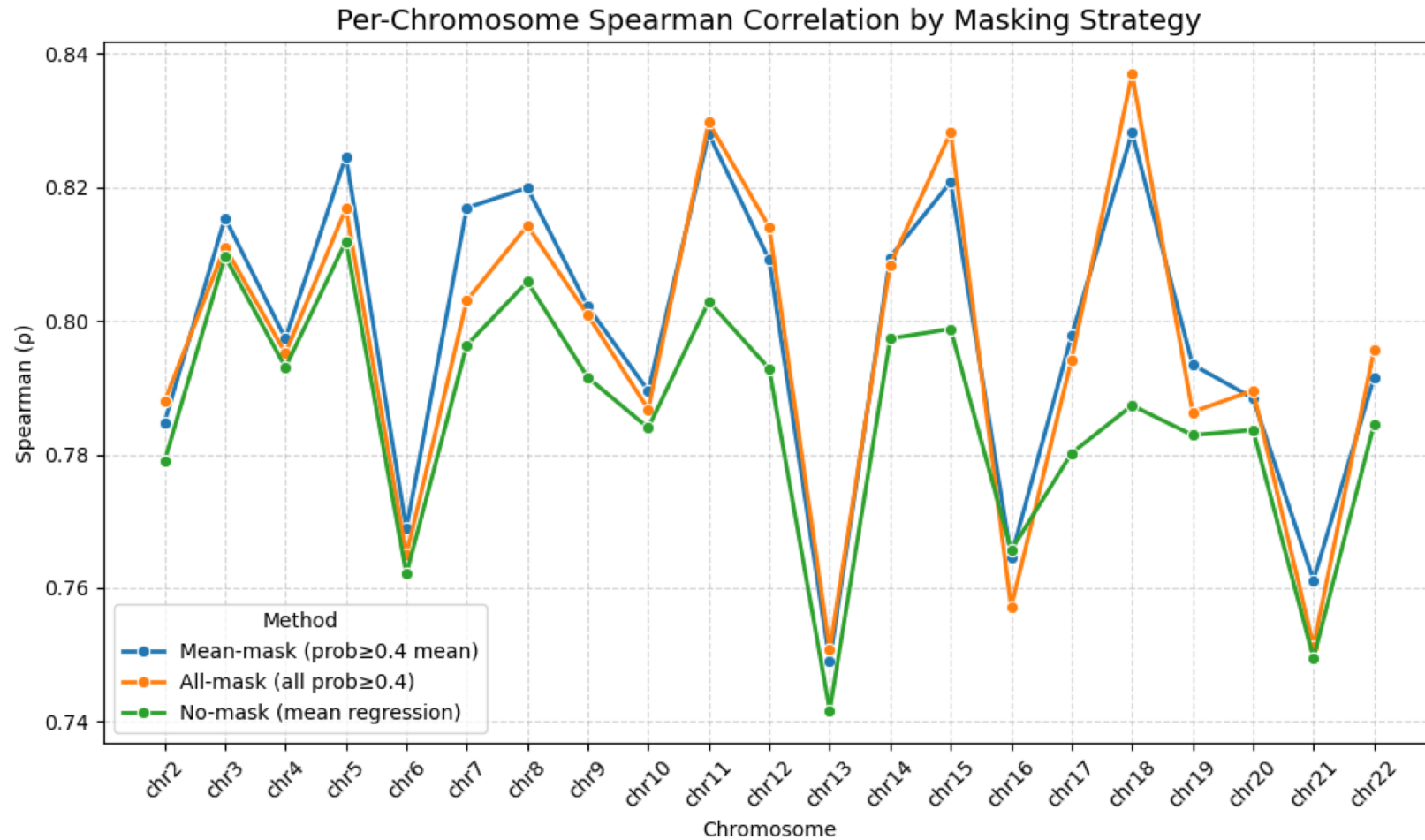
- leave-one-chromosome-out (LOCO). For each held-out chromosome, perform k-fold CV on the remaining chromosomes for both binary classification and regression.



Binary Threshold selection



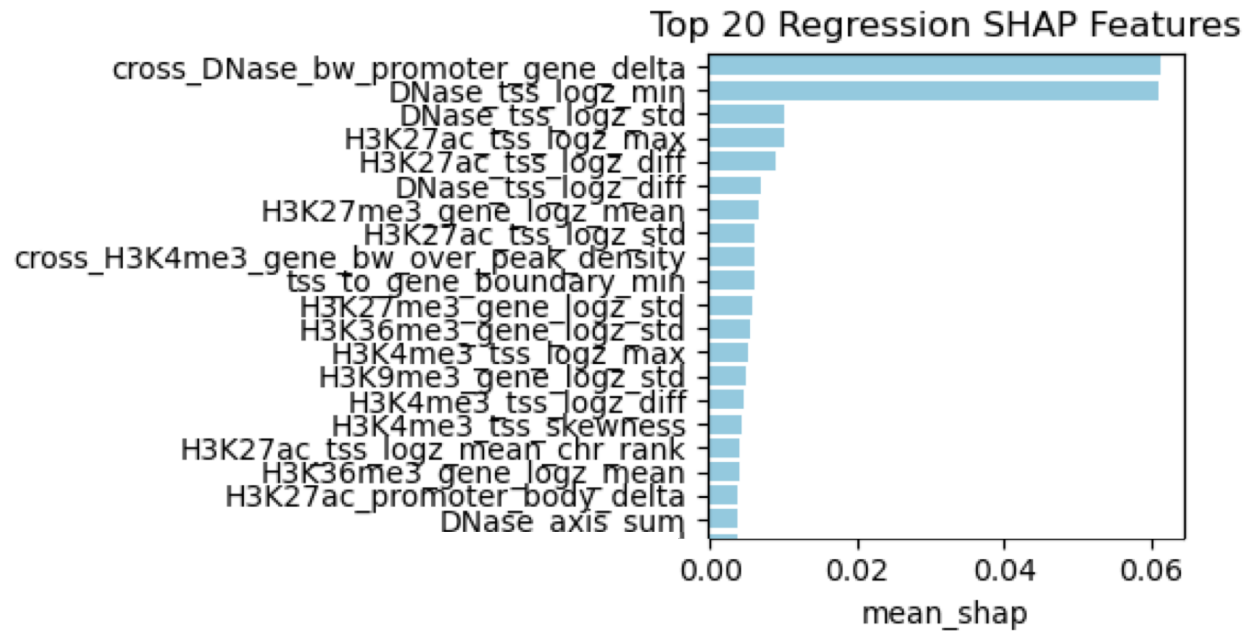
Binary Strategy selection



Prediction stacking

- Final predictions are **stacked** across the three setups.
 - X1+X2 pooled: mix X1+X2 for train/val
 - X1-only: train/val on X1
 - X2-only: train/val on X2

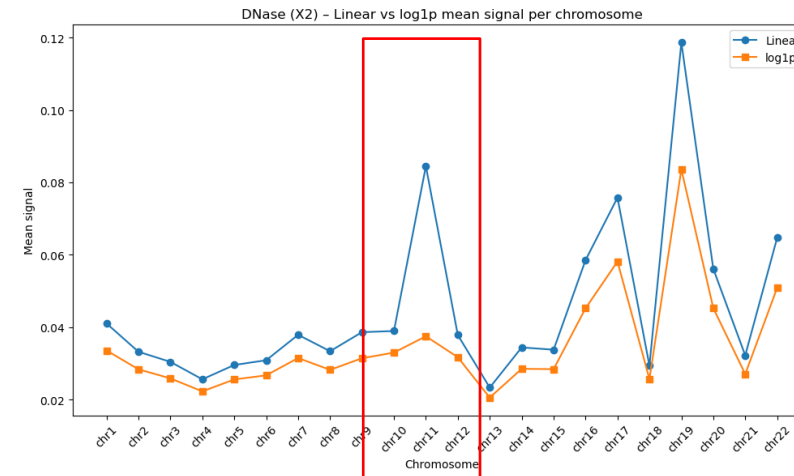
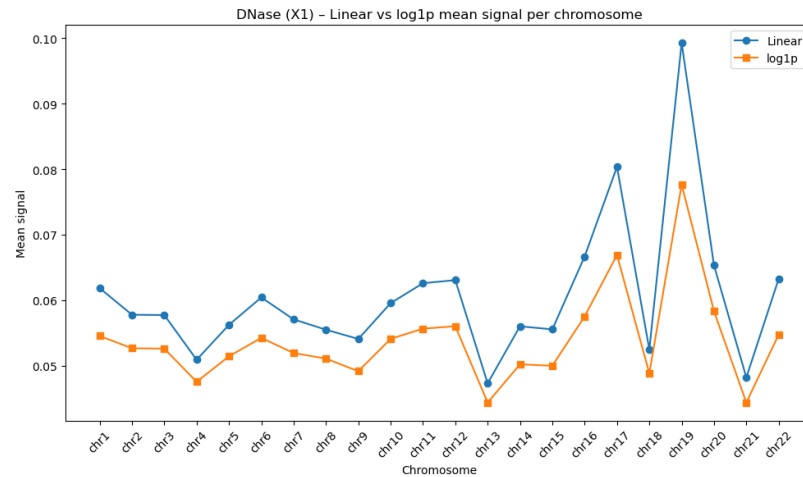
Result Interpretation



Chromatin accessibility is a crucial factor to predict the gene expression.

Question for you

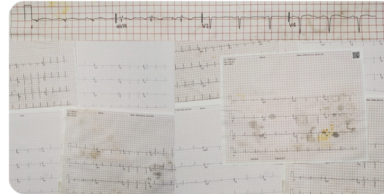
Why log1p and z-score have the best performance?



Competitions

PhysioNet - Digitization of ECG Images

Extract the ECG time-series data from scans and photographs of paper printouts of the ECGs.



CSIRO - Image2Biomass Prediction

Predict biomass using the provided pasture images



Nipah Binder Competition

Task: Protein Binding Target: Nipah Virus Glycoprotein G 1000 proteins experimentally validated

Design a protein capable of neutralizing the Nipah virus, a pathogen with up to 75% mortality rate and high pandemic potential, currently lacking effective treatments.

AI and ML for Microscopy, Hackathon

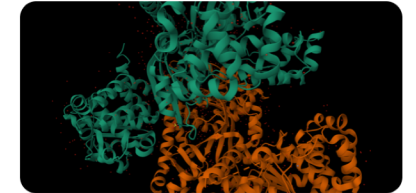
Machine Learning for Microscopy Hackathon 2

December 16–18, 2025 · Hybrid (In-person + Virtual)

[See the 2024 edition](#)

CAFA 6 Protein Function Prediction

Predict the biological function of a protein



2025 PETASE TOURNAMENT

A protein engineering competition employing high-throughput experimentation to benchmark model performance.

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